

Numerical Methods for Driven Damped Oscillator and Kepler Orbit Integrals

Fangyi Guo

Abstract

In this report, we present a comprehensive study of numerical methods applied to two classical physics problems: the driven damped harmonic oscillator and the Kepler orbit integral. We implement various algorithms, including the explicit Euler method, the 4th order Runge-Kutta method (RK4), Riemann sum, trapezoidal rule, and Simpson's rule, to solve ordinary differential equations and compute definite integrals. The results are validated against known analytic solutions, and convergence tests are conducted to analyze the accuracy of each method. We verify the expected order of global truncation errors as a function of time step for each algorithm. We also examine physical properties such as energy conservation and resonance in the oscillator problem and the divergence of the integral for non-elliptic orbits. Our results demonstrate the effectiveness of these numerical techniques and provide insights into their application in more complex scenarios.

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1 Introduction

The numerical study of dynamic systems is an important subject in computational physics. Many physical problems cannot be solved analytically, and approximate methods for simulation are useful in order to study such problems numerically. In this project, we investigate two canonical examples in classical physics: the driven damped harmonic oscillator and the orbit period problem in Kepler’s law of planetary motion. These problems are well-suited for our purpose of exploring numerical methods, as their analytic solutions are known, allowing us to study accuracy and convergence directly.

The oscillator problem models a mass attached to a spring subject to linear damping and period driving. The motion of the oscillator is governed by an inhomogeneous second-order linear ODE, which can be solved analytically. We apply various ODE solvers, including Euler’s method, Runge-Kutta, and the external solvers from the `scipy` package. Tracking the physical quantities such as energy of the problem and comparing with the analytic solutions allow us to study the strengths and limitations of these methods. The results are manifested and quantified as truncation errors as a function of timestep.

Kepler’s law solves the two-body motion in the classical regime, in particular giving the analytic result of the period of elliptic orbits. The formula is derived by computing a definite integral which only depends on the semi-major axis and the mass of the central body. By using Riemann sums, trapezoidal and Simpson’s rules, and their implementations in `scipy` library, we can compare with the analytic solution and conduct convergence test. It is also possible to examine the breakdown of the integral for non-elliptic orbit with eccentricity $e \geq 1$, where divergence is expected.

By studying the two simple examples, we provide a unified framework for implementation, validation and error analysis of numerical methods on differential equations and definite integrals. The generic strategy can be applied for more complicated problems where analytic solution might not be easily found, serving as a tool for estimation.

2 Analytic Review

For the purpose of comparing and validating the numerical methods, we briefly go over the analytic solutions in this section. Rather than giving a complete derivation, we list key results of each problem, which will be verified by numerical methods in the next section.

2.1 Driven Damped Oscillator

Consider the harmonic oscillator with mass m , spring constant k , damping constant c , sinusoidal drive force $F(t) = F_0 \cos(\omega t)$. The equation of motion is given by

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos(\omega t). \quad (2.1)$$

Let $\omega_0 = \sqrt{k/m}$ and $\beta = \frac{c}{2m}$, then the above equation can be written as

$$\ddot{x} + 2\beta\dot{x} + \omega^2 x = \frac{F_0}{m} \cos(\omega t). \quad (2.2)$$

The analytic solution has three regimes depending on the relation of β and ω_0 . The homogeneous solutions are

$$x_h(t) = \begin{cases} e^{-\beta t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t), & \beta < \omega_0 \\ (B_1 + B_2 t) e^{-\omega_0 t}, & \beta = \omega_0 \\ C_1 e^{r_+ t} + C_2 e^{r_- t}, & \beta > \omega_0 \end{cases} \quad (2.3)$$

where

$$\omega_d = \sqrt{\omega_0^2 - \beta^2}, \quad r_{\pm} = -\beta \pm \sqrt{\beta^2 - \omega_0^2}. \quad (2.4)$$

The constants can be determined by initial conditions. The particular solution to the equation of motion is given by

$$x_p(t) = X(\omega) \cos(\omega t - \phi(\omega)), \quad (2.5)$$

where

$$X(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\beta\omega)^2}}, \quad \phi(\omega) = \arctan \frac{2\beta\omega}{\omega_0^2 - \omega^2} \quad (2.6)$$

The full solution is thus given by $x(t) = x_h(t) + x_p(t)$. Given the initial conditions $x(0) = x_0, \dot{x}(0) = v_0$, we conclude the following:

Underdamped ($\beta < \omega_0$)

$$x(t) = e^{-\beta t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t) + X \cos(\omega t - \phi) \quad (2.7)$$

with

$$A_1 = x_0 - X \cos \phi, \quad A_2 = \frac{v_0 + \beta A_1 - \omega X \sin \phi}{\omega_d}. \quad (2.8)$$

Critically damped ($\beta = \omega_0$)

$$x(t) = (B_1 + B_2 t) e^{-\omega_0 t} + X \cos(\omega t - \phi) \quad (2.9)$$

with

$$B_1 = x_0 - X \cos \phi, \quad B_2 = v_0 + \beta (x_0 - X \cos \phi) - \omega X \sin \phi. \quad (2.10)$$

Overdamped ($\beta > \omega_0$)

$$x(t) = C_1 e^{r_+ t} + C_2 e^{r_- t} + X \cos(\omega t - \phi) \quad (2.11)$$

with

$$C_1 = \frac{(v_0 - \omega X \sin \phi) - r_- (x_0 - X \cos \phi)}{r_+ - r_-}, \quad C_2 = \frac{r_+ (x_0 - X \cos \phi) - (v_0 - \omega X \sin \phi)}{r_+ - r_-} \quad (2.12)$$

At the beginning, the system shows transient oscillations, which is described by the decaying exponential part of the solutions. After a typical time, the transient oscillation dies out and the solution settles into a steady-state oscillation at the driving frequency. We know from the analytic solution that the resonance appears for the underdamped case at

$$\omega_{\text{res}} = \sqrt{\omega_0^2 - 2\beta^2}, \quad (2.13)$$

where the amplitude of the particular solution $X(\omega)$ peaks. One can find the numerical value for this maximum amplitude.

The energy of the system is given by

$$E = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} k x^2. \quad (2.14)$$

Taking derivative and using the equation of motion, we have

$$\dot{E} = F_0 \cos(\omega t) \dot{x} - c \dot{x}^2. \quad (2.15)$$

One can check numerically that this identity holds. For simplicity, one can also check the special cases for undriven damped $F_0 = 0$ where $E(t)$ decreases monotonically, or for undriven undamped ($F_0 = 0, \beta = 0$) where $E(t)$ is constant.

2.2 Kepler Orbit Integral

Consider a particle of mass m in a central potential $V(r) = -\frac{\mu m}{r}$, then we have the equation of motion

$$\begin{aligned} m(\ddot{r} - r\dot{\theta}^2) &= -\frac{dV(r)}{dr}, \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) &= 0, \end{aligned} \quad (2.16)$$

where $\mu = GM$ is constant in our physical background. The second equation implies that the angular momentum, defined as

$$L = mr^2\dot{\theta}, \quad (2.17)$$

is constant. Let $u(\theta) = 1/r$, after some algebra the equation of motion can be simplified as

$$u'' + u = \frac{\mu m^2}{L^2}. \quad (2.18)$$

The ODE has the solution

$$u(\theta) = \frac{\mu m^2}{L^2} (1 + e \cos(\theta - \theta_0)), \quad (2.19)$$

and then the trajectory of the particle is given by

$$r(\theta) = \frac{p}{1 + e \cos(\theta - \theta_0)}, \quad p = \frac{L^2}{\mu m^2}. \quad (2.20)$$

This expression defines conic curves for the trajectory, and we will focus on ellipse case where the eccentricity $e < 1$. Using the definition of the angular momentum, we can compute the period of the orbit

$$I = \int_0^T dt = \frac{m}{L} \int_0^{2\pi} r(\theta)^2 d\theta = \frac{p^2 m}{L} \int_0^{2\pi} \frac{d\theta}{(1 + e \cos \theta)^2}. \quad (2.21)$$

The last inequality involves a constant shift $\theta \rightarrow \theta + \theta_0$, and hence this integral is independent of θ_0 . Analytically, the integral evaluates to be

$$\int_0^{2\pi} \frac{d\theta}{(1 + e \cos \theta)^2} = \frac{2\pi}{(1 - e^2)^{3/2}}. \quad (2.22)$$

For the simulation purpose, one can choose the parametrization using a, e, μ, m , which then gives p, L . Using the definition of p and the length of the semi-major axis $a = p/(1 - e^2)$, we derive Kepler's third law,

$$T = 2\pi \sqrt{\frac{a^3}{\mu}}, \quad (2.23)$$

which can be a physical quantity to compare with the simulation result.

Note that the integral diverges for $e \geq 1$, which corresponds to parabolic and hyperbolic orbits. This can also be checked numerically.

3 Numerical Methods

In this section, we will briefly introduce the numerical methods we use for solving the problems. In each case, we implemented multiple algorithms of varying order, together with external solvers from the `scipy` library, in order to validate results and study convergence.

3.1 ODE Solvers

For the driven damped oscillator, we solve the second-order ordinary differential equation [2.1](#)

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos(\omega t). \quad (3.1)$$

This second-order ODE can be written as a system of two first-order ODEs, on which we will apply the explicit Euler method, the 4th order Runge-Kutta method (RK4) and the SciPy IVP solver.

The explicit Euler method is the simplest time-stepping scheme for solving ordinary differential equations (ODEs). Starting from an initial state x_0, v_0 , the method advances the solution by evaluating the right-hand side of the ODE at the current step

$$y_{n+1} = y_n + hf(t_n, y_n), \quad (3.2)$$

where y is the state vector, h is the timestep, and f is the function defining the ODE. For our problem, we define the state vector as $y = \begin{pmatrix} x \\ v \end{pmatrix}$, and the function f is given by

$$f(t, y) = \begin{pmatrix} v \\ -\frac{c}{m}v - \frac{k}{m}x + \frac{F_0}{m} \cos(\omega t) \end{pmatrix}. \quad (3.3)$$

The method is first-order accurate, meaning the local truncation error per step is at order $\mathcal{O}(h^2)$, and the global error over N steps is thus $\mathcal{O}(h)$.

The 4th order Runge-Kutta method (RK4) is a widely used explicit method for solving ODEs, known for its balance between accuracy and computational efficiency. The RK4 method advances the solution by computing four intermediate slopes (k-values) at each timestep, which are then combined to produce a weighted average slope. The update formula is given by

$$\begin{aligned} k_1 &= hf(t_n, y_n), \\ k_2 &= hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right), \\ k_3 &= hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right), \\ k_4 &= hf(t_n + h, y_n + k_3), \\ y_{n+1} &= y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \quad (3.4)$$

The RK4 method has more accuracy than the explicit Euler method. In fact, it is fourth-order accurate, with the local truncation error per step at order $\mathcal{O}(h^5)$. Over N steps, the global error is $\mathcal{O}(h^4)$.

For comparison, we also use the `solve_ivp` function from the SciPy library, which provides a variety of ODE solvers. We choose the ‘RK45’ method, which is an explicit Runge-Kutta method of order 5(4). By examining the results from these three methods, we can validate our implementations and study their convergence properties.

3.2 Numerical Integration

For the Kepler orbit problem, we study the definite integral

$$I = \int_0^{2\pi} \frac{d\theta}{(1 + e \cos \theta)^2}, \quad (3.5)$$

which gives the period of the orbit. We implement the Riemann sum, trapezoidal rule, Simpson’s rule, and use the `quad` function from the SciPy library to compute the integral numerically.

The Riemann sum is the most straightforward method for numerical integration. It approximates the integral by dividing the integration interval into N subintervals of equal width $h = \frac{b-a}{N}$, with a, b the integration bounds, and summing the function values at specific points within each subinterval. We use the midpoint rule in our implementation, where the function is evaluated at the midpoint of each subinterval. The Riemann sum is given by

$$I \approx \sum_{i=0}^{N-1} h f \left(a + \left(i + \frac{1}{2} \right) h \right). \quad (3.6)$$

For each subinterval, due to the selection of the midpoint, the second-order term in the Taylor expansion cancels out, and the local truncation error is $\mathcal{O}(h^3)$, leading to a global error of $\mathcal{O}(h^2)$ over N subintervals.

The trapezoidal rule improves upon the Riemann sum by approximating the area under the curve as a series of trapezoids rather than rectangles. The integral is approximated as

$$I \approx \frac{h}{2} \left(f(a) + 2 \sum_{i=1}^{N-1} f(a + ih) + f(b) \right). \quad (3.7)$$

Again, by taking the average of the function values at the endpoints of each subinterval, the trapezoidal rule achieves a local truncation error of $\mathcal{O}(h^3)$ per subinterval, resulting in a global error of $\mathcal{O}(h^2)$ over N subintervals.

Simpson’s rule provides a higher-order approximation by fitting a quadratic polynomial to each pair of subintervals. The integral is approximated as

$$I \approx \frac{h}{3} \left(f(a) + 4 \sum_{i=1, \text{ odd}}^{N-1} f(a + ih) + 2 \sum_{i=2, \text{ even}}^{N-2} f(a + ih) + f(b) \right), \quad (3.8)$$

where N must be even. By construction, it turns out that the local truncation error is $\mathcal{O}(h^5)$ per pair of subintervals, which gives a global error of $\mathcal{O}(h^4)$ over N subintervals.

We also use the `trap` and `simps` functions from the SciPy library, which implement the trapezoidal and Simpson’s rules, respectively. By comparing the results from our implementations with those from SciPy, we can validate our methods and study their convergence properties.

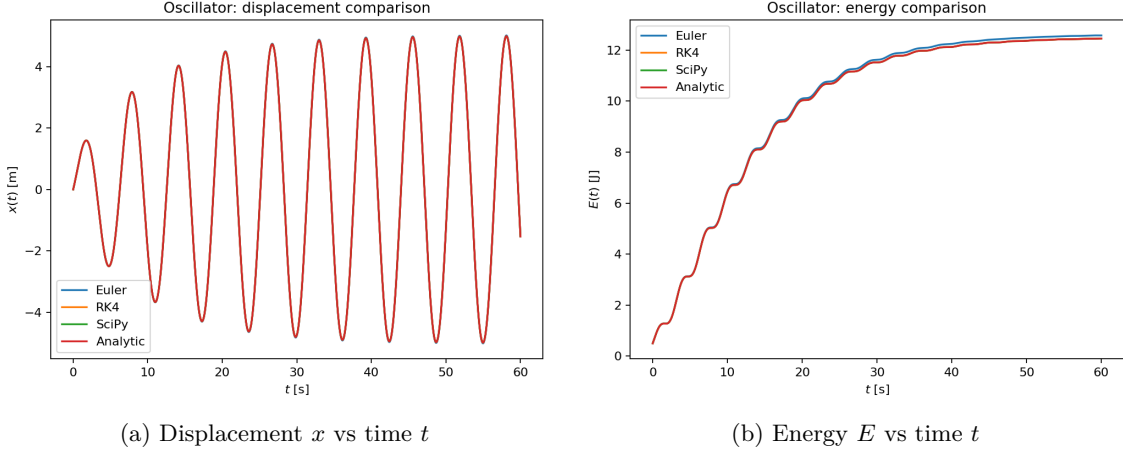


Figure 1: Comparison of Euler, RK4 and SciPy solution with analytic solution for position (left) and energy (right). Parameters used are $m = 1$, $k = 1$, $c = 0.2$, $\omega = 1$, $F_0 = 1$, $x_0 = 0$, $v_0 = 1$, $dt = 0.001$, $t_{\max} = 60$.

4 Results and Discussion

In this section, we present the results obtained from the numerical methods applied to the driven damped oscillator and the Kepler orbit integral. We analyze the accuracy and convergence of each method by comparing numerical results with the known analytic solutions. We also check the physical properties such as energy conservation and resonance behavior in the oscillator problem, and the divergence in the integral for non-elliptic orbits.

4.1 Driven Damped Oscillator

For the driven damped oscillator, we implement the explicit Euler method, the 4th order Runge-Kutta method (RK4), and the SciPy IVP solver to solve the equation of motion (3.1). We compare the numerical results with the analytic solution in terms of displacement $x(t)$ and energy $E(t)$. We also study the resonance behavior by varying the driving frequency ω .

4.1.1 Time-domain Behavior

For a typical set of parameters $m = 1$, $k = 1$, $c = 0.2$, $\omega = 1$, $F_0 = 1$, $x_0 = 0$, $v_0 = 1$, we simulate the system up to $t_{\max} = 60$ with a timestep $dt = 0.001$. The results from the three numerical methods are compared with the analytic solution in Figure 1. We observe that all three methods capture the transient and steady-state behavior of the oscillator accurately. For the displacement $x(t)$, all three methods show good agreement with the analytic solution. However, by tracking the energy $E(t)$, the RK4 and SciPy methods show better agreement with the analytic solution in terms of energy conservation over long times. The Euler method exhibits noticeable deviations in energy as time increases, indicating its higher global error accumulation.

To capture the truncation error of the methods, we perform simulations with varying timesteps dt ranging from 5×10^{-4} to 5×10^{-2} . We compute the root mean square (RMS) error of the displacement compared to the analytic solution and fit the data linearly as a function of time step on a log-log scale. The results are plotted in Figure 2. The slopes of the fitted lines indicate the

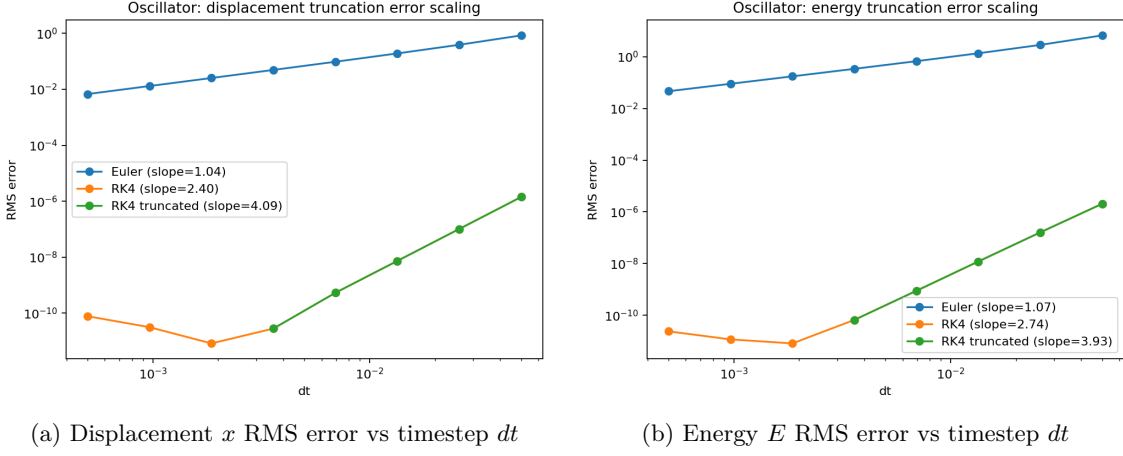


Figure 2: Convergence of Euler and RK4 methods for displacement x (left) and energy E (right). The dots indicate the simulation errors at time steps ranging from 5×10^{-4} to 5×10^{-2} . The slopes of the fitted lines indicate the order of convergence. Parameters used are $m = 1$, $k = 1$, $c = 0.2$, $\omega = 1$, $F_0 = 1$, $x_0 = 0$, $v_0 = 1$, $t_{\max} = 60$.

order of convergence for each method. Note that for the RK4 method, we also fit the slope using only the smaller time steps.

The results for the displacement x and energy E are consistent. As is shown in the plots, the slopes for the Euler method and truncated RK4 are approximately 1 and 4. This give first-order convergence for the Euler method and fourth-order convergence for the RK4 method as expected. This result is consistent with the theoretical analysis of the global truncation errors in Section 3.1. We can infer from the plots that the linear fits goes on as time step increases, confirming the global error behavior. For RK4, there exists a step time scale between 10^{-3} and 10^{-2} where the curve flattens out. This is likely due to the accumulation of round-off errors at very small time steps, which can dominate the truncation error and lead to a U-shaped curve.

Moreover, the absolute errors for RK4 are significantly smaller than those for Euler at the same timestep, demonstrating the better accuracy of the higher-order method. The detailed numerical values of the fitted slopes are printed in the console output.

4.1.2 Resonance Behavior

To study the resonance behavior, we vary the driving frequency ω from 0.3 to 2.0 and simulate the system for each frequency using the RK4 method. We use the same parameters as before, and truncate the first half of the simulation time to avoid transient effects. To guarantee accuracy for all frequencies, we specify the number of cycles to be simulated at each frequency and the number of sampling points per cycles.

We measure the amplitude of the steady-state oscillation after the transient behavior has died out. The results are plotted in Figure 3. The numerical results fit perfectly with the analytic anticipation at all frequencies within the simulation range. We observe a clear resonance peak near the theoretical resonance frequency $\omega_{\text{res}} = \sqrt{\omega_0^2 - 2\beta^2} \approx 0.99$. The numerical results match well with the analytic prediction, validating the accuracy of the RK4 method in capturing the resonance behavior of the driven damped oscillator.

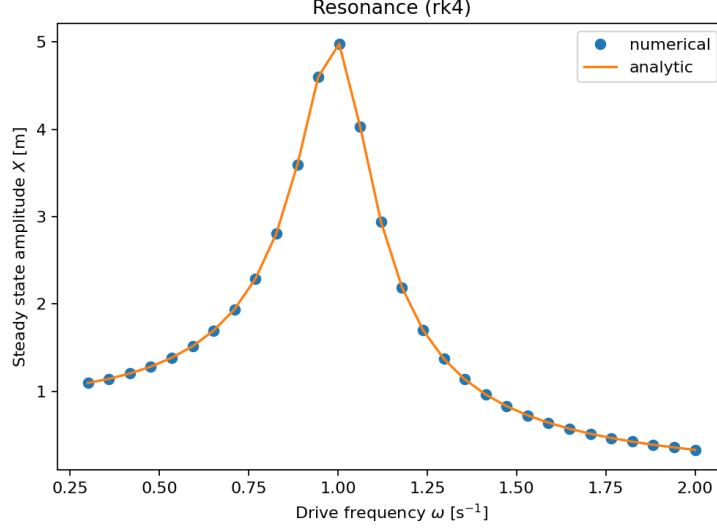


Figure 3: Resonance behavior of the driven damped oscillator. The amplitude of the steady-state oscillation is plotted against the driving frequency ω . The numerical results (dots) are aligned with the analytic prediction (line). Parameters used are $m = 1$, $k = 1$, $c = 0.2$, $F_0 = 1$, $x_0 = 0$, $v_0 = 1$, $dt = 0.001$, $t_{\max} = 60$. We also specify 60 cycles and 200 sampling points per cycle for each frequency.

4.2 Orbit

The definite integral for the Kepler orbit problem is given by (3.5). We divide the integration interval into N subintervals

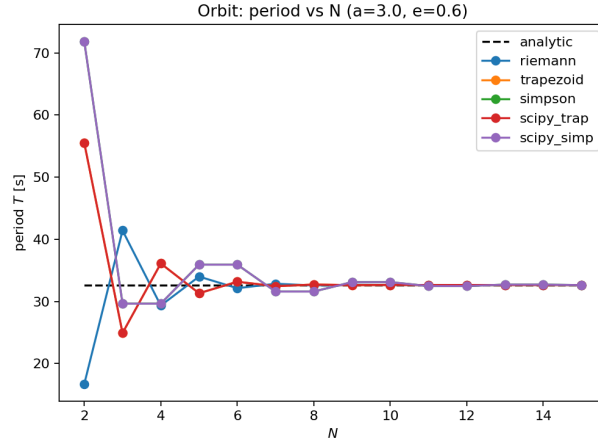
$$h = \frac{2\pi}{N}, \quad (4.1)$$

and use the algorithms including Riemann sum, trapezoidal rule and Simpson's rule. As a reference, the `trap` and `simps` functions from the SciPy library are also used. We compare the numerical results with the analytic solution in terms of the orbit period T . We also study the convergence of each method by varying the number of subintervals N . Finally, we check the divergence of the integral for eccentricity $e \geq 1$.

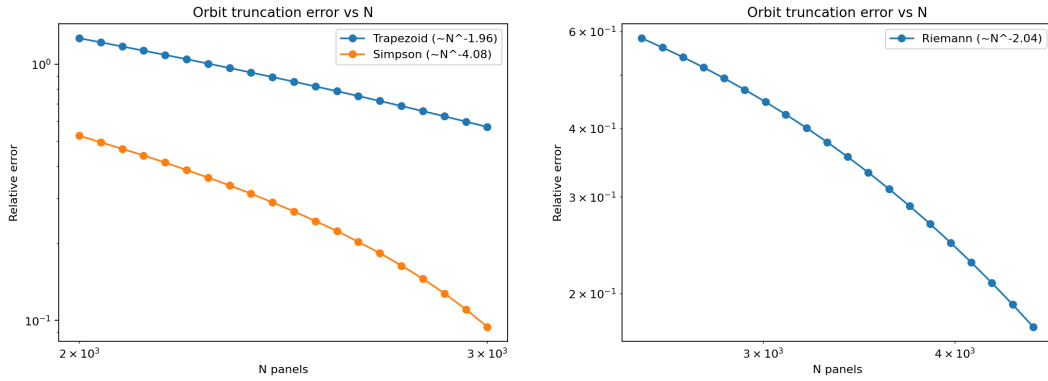
4.2.1 Period Calculation

For a typical set of parameters $a = 3$, $e = 0.6$, $\mu = 1$, we compute the orbit period T using the numerical integration methods with $N = 1000$ subintervals. At this high number of subintervals, all methods yield results that are very close to the analytic solution $T = 2\pi\sqrt{a^3/\mu} \approx 32.648$, with relative errors lower than 10^{-16} , the machine precision for double-precision floating-point numbers. Precise values and relative errors with more digits for each method are printed in the console output. The results also show that the SciPy implementations of the trapezoidal and Simpson's rules yield results identical to our implementations up to machine precision.

The extreme accuracy at this high N makes it difficult to distinguish the performance of different methods. To better illustrate the differences, we perform simulations at small N 's ranging from 2 to 15. The results of the various methods are compared with the analytic solution from Kepler's



(a) Period integral T vs number of subintervals N



(b) Convergence of Riemann sum, trapezoidal rule and Simpson's rule

Figure 4: Comparison of numerical methods for the Kepler orbit integral. (Top) The computed orbit period T is plotted against the number of subintervals N . The numerical results from Riemann sum, trapezoidal rule, Simpson's rule, and their SciPy counterparts are compared with the analytic solution (dashed line). Parameters used are $a = 3$, $e = 0.6$, $\mu = 1$. (Bottom) The absolute errors of the integral for Riemann sum, trapezoidal rule and Simpson's rule are plotted against the number of subintervals N on a log-log scale. The slopes of the fitted lines indicate the order of convergence. The eccentricity used is $e = 1 - \epsilon$ with $\epsilon = 10^{-7}$.

third law in Figure 4(a). We observe that all numerical methods show clear discrepancies from the analytic solution for $N < 10$. As N increases, the numerical results converge towards the analytic solution. The Simpson's rule and its SciPy counterpart show the fastest convergence, followed by the trapezoidal rule and its SciPy counterpart, and finally the Riemann sum. This behavior is consistent with the theoretical analysis of the global truncation errors in Section 3.2.

To better characterize the global truncation errors, we calculate the absolute errors of the integral for each non-SciPy method and plot them against the number of subintervals N on a log-log scale. However, the definite integral we are considering is not complicated enough to show the full convergence behavior within a reasonable range of N . To address this, we set the eccentricity to be very close to 1, specifically $e = 1 - \epsilon$ with $\epsilon = 10^{-7}$. This choice makes the integrand sharply peaked, effectively increasing the complexity of the integral and allowing us to observe the convergence behavior more clearly. Moreover, instead of integrating from 0 to 2π , we integrate

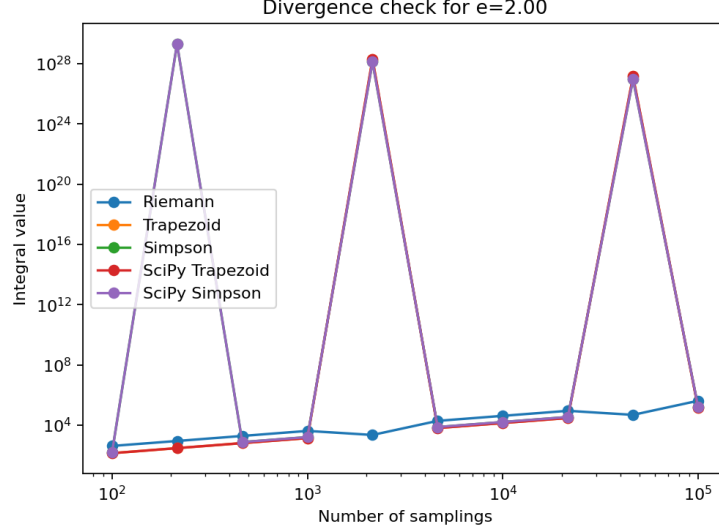


Figure 5: Divergence check of the integral I for eccentricity $e = 2$. The numerical results from Riemann sum, trapezoidal rule, Simpson’s rule, and their SciPy counterparts are plotted against the number of subintervals N . Parameters used are $a = 3$, $\mu = 1$.

from 0 to π and multiply the result by 2, which is equivalent due to the symmetry of the integrand. This adjustment helps to avoid numerical issues near the singularity at $\theta = \pi$ when e is very close to 1.

The results are shown in Figure 4(b). The logarithm of the data points are fitted linearly, so that the slopes of the fitted lines indicate the order of convergence for each method. Since different methods have different convergence rates, we use different ranges of N for the linear fits to avoid the influence of higher-order terms in the truncation error. The resulting slopes shown in the plots are approximately -2, -2, and -4 for the Riemann sum, trapezoidal rule and Simpson’s rule, respectively. This corresponds to $\mathcal{O}(h^2)$, $\mathcal{O}(h^2)$ and $\mathcal{O}(h^4)$ global convergence according to (4.1), which is consistent with the theoretical analysis in Section 3.2. The detailed numerical values of the fitted slopes are printed in the console output.

Besides the expected truncation error behavior, we also observe that the relative errors concave down as N increases. This indicates that the linear fits will not hold for large N , as the higher-order terms in the truncation error become significant. This is particularly evident for the Simpson’s rule, where its curve bends down quickly for $N > 3000$.

4.2.2 Divergence Check

A special feature of the integral (3.5) is that it diverges for eccentricity $e \geq 1$. This corresponds to parabolic and hyperbolic orbits, where the particle escapes to infinity. In fact, the denominator of the integrand (3.5) becomes zero at $\cos \theta = -\frac{1}{e}$ for $e \geq 1$, leading to a singularity in the integrand and hence divergence of the integral.

We can check this divergence numerically by setting $e = 2$ and computing the integral using the various numerical methods with increasing number of subintervals N . The results are shown in Figure 5, which have a very large order of magnitude. We observe that for all methods, the

computed values of the integral increase without bound as N increases, implying a clear trend of divergence.

The plot also exhibits several unnatural peaks, which are likely due to the situation that the grid lands very close to the pole and hence the integrand becomes extremely large. Noticeably, these peaks only appear for trapezoidal rule, Simpson's rule and their SciPy counterparts, but not for the Riemann sum. This is because the Riemann sum samples the function at the midpoints of each subinterval, which avoids directly sampling at the singularity. In contrast, the trapezoidal and Simpson's rules sample at the endpoints of the subintervals, which can include the singularity if it falls on a grid point.

5 Conclusion

In this project, we have explored numerical methods for solving ordinary differential equations and definite integrals through the study of two classical physics problems: the driven damped harmonic oscillator and the Kepler orbit integral. By implementing multiple algorithms of varying order, including the explicit Euler method, the 4th order Runge-Kutta method (RK4), Riemann sum, trapezoidal rule, and Simpson's rule, we have shown consistency with the SciPy functions and validated our implementations against known analytic solutions. We have also conducted convergence tests explicitly for each method, studying and verifying the expected order of global truncation errors. Furthermore, we have examined the physical properties of the problems, such as energy conservation and resonance behavior in the oscillator problem, and the divergence of the integral for non-elliptic orbits, where our numerical results show excellent agreement with theoretical expectations.

In the future, the framework established in this project can be extended to more complex problems in physics, where analytic solutions may not be readily available. The insights gained from the convergence analysis and error estimation can guide the selection of appropriate numerical methods for various applications, ensuring both accuracy and computational efficiency.